Statistical Detection of Cyberattacks Using Network Traffic Data

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Course: Probability and Statistics

Submission note: This report is a ready-to-complete project report. Add CIC-IDS2017 CSV data to the data folder and run the included R script. The script generates the actual results; this report intentionally does not invent numerical findings.

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Abstract

Network intrusion detection can be studied as a statistical classification problem by examining whether measurable properties of network flows differ between benign and malicious traffic. This project uses the CIC-IDS2017 network-flow dataset and R to summarize traffic features, visualize distributions, test differences between traffic classes, and estimate the probability that a flow is malicious using logistic regression. Performance is evaluated on a held-out test sample with a confusion matrix, accuracy, sensitivity, specificity, precision, and ROC-AUC when calculable. Numerical findings are generated by the included R script after the dataset is added; no results are invented in advance.

1. Introduction

Computer networks generate large volumes of flow records. Each record can contain measurements such as flow duration, packet counts, byte counts, and packet-rate statistics. Benign activity and attacks may differ in these measurements, although no single feature perfectly identifies every attack. Probability and statistics provide tools for describing variation, testing group differences, quantifying uncertainty, and building interpretable predictive models.

2. Aim and Objectives

Aim: To investigate whether statistical properties of network traffic can help distinguish benign traffic from cyberattacks.

Objectives:

1. Summarize selected network-flow features using descriptive statistics.
2. Compare feature distributions for benign and malicious traffic.
3. Test whether selected features differ between the two groups.
4. Fit a logistic regression model to estimate attack probability.
5. Evaluate classification performance on held-out observations.
6. Discuss limitations, uncertainty, and possible sources of bias.

3. Dataset

The proposed source is CIC-IDS2017, published by the Canadian Institute for Cybersecurity. It contains labeled benign and attack traffic represented as network-flow records. Obtain CSV flow files from the official page: https://www.unb.ca/cic/datasets/ids-2017.html

The analysis script detects a label column named `Label` or a similar common variant. Labels containing `BENIGN` are treated as benign; other non-empty labels are treated as attacks. Confirm that this rule matches the downloaded file before interpreting results.

Dataset documentation to record after download:

- File(s) used: [complete after download]
- Number of records read: [generated by R script]

• Number retained after cleaning/sampling: [generated by R script]

• Benign records: [generated by R script]

• Attack records: [generated by R script]

• Features used: [generated by R script]

4. Statistical Methodology

4.1 Data preparation

The script combines CSV files from the `data/` directory, standardizes column names for matching, identifies the target label, converts selected features to numeric values, replaces infinite values with missing values, removes unusable predictors, and samples a manageable number of records. The fixed random seed makes the sampling and train/test split reproducible.

4.2 Descriptive statistics

For each selected numeric feature, calculate:

• Mean: $\bar{x}=\frac{1}{n}\sum_{i=1}^{n}x_i$

• Sample variance: $s^2=\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$

• Standard deviation: $s=\sqrt{s^2}$

• Median and quartiles, which are less sensitive to extreme values.

Histograms and boxplots help identify skewness, outliers, and class differences.

4.3 Hypothesis testing

For selected features, compare benign and attack groups.

For the Welch two-sample t-test:

• Null hypothesis H_0: the two population means are equal.

• Alternative hypothesis H_1: the two population means differ.

• Significance level: $\alpha=0.05$.

Welch's test does not require equal population variances. A Wilcoxon rank-sum test is also included as a distribution/rank-based comparison. A small p-value indicates evidence against the relevant null hypothesis, but does not by itself show that a difference is practically important. Multiple feature tests increase the chance of false positives; interpret the results cautiously.

4.4 Logistic regression

Let $Y=1$ indicate an attack and $Y=0$ indicate benign traffic. Logistic regression models

$$
P(Y=1\mid X)=\frac{1}{1+e^{-(\beta_0+\beta_1X_1+\cdots+\beta_kX_k)}}.
$$

The coefficients describe the association between predictors and log-odds of an attack, conditional on the other included predictors. They should not automatically be interpreted causally. The model is fitted on training data and evaluated on held-out data.

4.5 Evaluation metrics

• Accuracy = (TP + TN) / (TP + TN + FP + FN)

• Sensitivity/recall = TP / (TP + FN)

• Specificity = TN / (TN + FP)

• Precision = TP / (TP + FP)

• False-positive rate = FP / (FP + TN)

The script calculates approximate 95% Wilson confidence intervals for accuracy, sensitivity, and specificity. ROC-AUC is calculated from predicted probabilities if both classes are present.

5. Results

Run `source("R/analysis.R")` after adding dataset CSV files to `data/`. The script creates:

• `results/dataset_summary.txt`

• `results/descriptive_statistics.csv`

• `results/hypothesis_tests.csv`

• `results/model_metrics.txt`

• `results/model_coefficients.csv`

• figures under `plots/`

Fill the following summary using the generated output:

1. **Dataset composition:** [insert observed counts]

2. **Largest descriptive differences:** [identify features from the output]

3. **Hypothesis-test results:** [report test statistic/p-value and interpret]

4. **Model performance:** [report test-set metrics and confidence intervals]

5. **Most informative predictors:** [describe coefficient estimates and uncertainty, where available]

Do not report placeholders as findings. All results must be based on the dataset actually analyzed.

6. Discussion

Differences in flow features can provide evidence that malicious and benign traffic have different statistical patterns. However, large datasets can make small differences statistically significant, so effect sizes and plots matter alongside p-values. Class imbalance can make accuracy misleading; sensitivity, specificity, precision, and the confusion matrix should be considered together. Logistic regression is interpretable but may miss nonlinear relationships and interactions.

A random row-level split may put closely related flows in both training and test sets, potentially inflating apparent performance. A stronger follow-up analysis would split by capture day or file, if the data layout supports it, and validate the model on an independent environment.

7. Limitations

1. Results depend on the selected CIC-IDS2017 files and their labels.

2. The dataset is a benchmark and may not represent current real-world traffic.

3. Duplicate or highly correlated records can affect evaluation.

4. Statistical association does not prove that a feature causes an attack.

5. Logistic regression may be affected by multicollinearity and class imbalance.

6. Random splitting may overestimate performance on unseen networks.

7. Statistical tests across many features require care about multiple comparisons.

8. Conclusion

This project provides a reproducible statistical workflow for investigating network-traffic-based cyberattack detection. It combines descriptive statistics, visual exploration, hypothesis testing, and probabilistic classification in R. The final conclusion should be written only after running the analysis and reviewing the generated tables and plots. The key question is not merely whether the model classifies records, but whether the evidence supports meaningful, reliable separation between benign and malicious traffic.

References

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